

10. THE PROFILE IN ARBITRARY CHARACTERISTIC, AND THE CHARACTERISTIC-ZERO PICTURE

Addendum (27 September 2026). Sections 1–9 treat $G = \mathbb{Z}/p^n$ in characteristic p . This section records what the definition of the profile gives for an arbitrary finite group G over an arbitrary field k , and what it looks like in characteristic 0 (more generally, in characteristic prime to $|G|$), where the mechanisms are entirely different: instead of Artin–Schreier layers, the profile is governed by Sylow divisibility (Theorem 10.10) and by the roots-of-unity defect $\mathrm{ed}_k(G) - \mathrm{ed}_{k(\mu_e)}(G)$ (Corollary 10.5). Statements carry the tags of Section 1; in addition, §10.6 says explicitly which statements are rewrites of results in the literature and which are new. As in the rest of the note, literature is quoted without re-verification and the corresponding statements are tagged [cited].

10.1. Conventions. k is any field, G a finite group (constant group scheme over k), $F \supseteq k$ a field, $\tau: T \rightarrow \mathrm{Spec} F$ a G -torsor. The classifying homomorphism $\varphi_\tau: \Gamma_F \rightarrow G$ is defined up to conjugation, its image $H = \mathrm{im} \varphi_\tau$ up to conjugacy, and $T \cong \coprod_{G/H} \mathrm{Spec} E$ with E/F Galois with group H (the *splitting field* of τ). We use, in every characteristic: (B1) $\mathrm{ed}_k(\tau \otimes_F F') \leq \mathrm{ed}_k(\tau)$; (B2) $\mathrm{ed}_k(\tau) \geq \mathrm{ed}_{k'}(\tau)$ for $k \subseteq k' \subseteq F$; (B3) $\mathrm{ed}_k(\tau) = 0$ iff τ descends to the algebraic closure of k in F (for k algebraically closed: iff τ is split; for general k this is *not* “descends to k ”: $k = \mathbb{Q}$, $F = \mathbb{Q}(i)(t)$, $\tau = F(\sqrt{1+i})$ has $\mathrm{ed}_{\mathbb{Q}} = 0$ but is not $F(\sqrt{a})$ with $a \in \mathbb{Q}$, since $N_{\mathbb{Q}(i)/\mathbb{Q}}(1+i) = 2$ is not a square); (B4) Lemma 4.1(i), purely transcendental base change does not change ed_k . For a prime q , $\mathrm{ed}_k(\tau; q) = \min\{\mathrm{ed}_k(\tau \otimes F') : F'/F \text{ finite separable, } q \nmid [F' : F]\}$ and $\mathrm{ed}_k(G; q) = \sup_\tau \mathrm{ed}_k(\tau; q)$ [25, (2.5)]. The generic torsor $\tau_{\mathrm{gen}}: \mathrm{Spec} L \rightarrow \mathrm{Spec} K$, $L = k(V)$, $K = L^G$, is taken for a faithful representation V (or any faithful generically free G -variety with $k(V)/k$ regular); $\mathrm{ed}_k(\tau_{\mathrm{gen}}) = \mathrm{ed}_k(G)$ [18, 23]. The profile $\mathrm{ed}_k^{[d]}(\tau)$ and the jump degrees $d_j(\tau)$ are as in Definition 3.1.

10.2. Formal properties, valid in every characteristic.

Proposition 10.1 (shape). *Let τ be a G -torsor over $F \supseteq k$.*

- (a) $d \mapsto \mathrm{ed}_k^{[d]}(\tau)$ is non-increasing with values in $\{0, \dots, \mathrm{ed}_k(\tau)\}$, $\mathrm{ed}_k^{[1]}(\tau) = \mathrm{ed}_k(\tau)$, and $d_{\mathrm{ed}(\tau)} = 1 \leq d_{\mathrm{ed}(\tau)-1} \leq \dots \leq d_0 \leq |\mathrm{im} \varphi_\tau|$.
- (b) If $[F' : F] = e$ and $d \geq 1$ then $\mathrm{ed}_k^{[de]}(\tau) \leq \mathrm{ed}_k^{[d]}(\tau \otimes F') \leq \mathrm{ed}_k^{[d]}(\tau)$; hence $d_j(\tau) \leq [F' : F] d_j(\tau \otimes F')$ for every finite F'/F and every j .
- (c) $\mathrm{ed}_k^{[d]}(\tau) = \min\{j : d_j(\tau) \leq d\}$: the jump degrees determine the profile.
- (d) If $k \subseteq k' \subseteq F$ then $\mathrm{ed}_{k'}^{[d]}(\tau) \leq \mathrm{ed}_k^{[d]}(\tau)$ for all d .
- (e) $\mathrm{ed}_k^{[d]}(\tau) = 0$ iff some F'/F of degree $\leq d$ makes $\tau \otimes F'$ descend to the algebraic closure of k in F' ; for k algebraically closed, iff τ is split by an extension of degree $\leq d$.

[proved]

Proof. (a) Over the splitting field E , $[E : F] = |H|$, the torsor is split, so $d_0 \leq |H|$; the rest is (B1). (b) is Proposition 3.2(b) verbatim; if $[F'' : F] = d_j(\tau \otimes F')$ realises $\mathrm{ed}(\tau \otimes F'') \leq j$ then $[F'' : F] = [F' : F] d_j(\tau \otimes F')$. (c) is a restatement, (d) is (B2) applied to each $\tau \otimes F'$, (e) is (B3). $\square$

Proposition 10.2 (the last jump degree). *Let $H = \mathrm{im} \varphi_\tau$ and E the splitting field of τ .*

- (a) $d_0(\tau) \leq \min\{[F' : F] : \tau \otimes F' \text{ split}\} = |H|$, with equality if k is algebraically closed.
- (b) Let k be perfect and E/k regular. If $[F' : F] < \infty$ and $\tau \otimes F'$ descends to the algebraic closure k_1 of k in F' , then $[F' : F] \geq |H| \cdot [k_1 : k]$. In particular $d_0(\tau) = |H|$.

- (c) For τ_{gen} over a perfect k : $d_0(\tau_{\text{gen}}) = |G|$, and $\text{ed}_k(\tau_{\text{gen}} \otimes F') = 0$ forces $[F' : K] \geq |G| [k_1 : k]$.

[proved]

Proof. (a) $T \otimes_F F'$ is split iff E embeds F -linearly into F' , which forces $[F' : F] \geq |H|$; $F' = E$ works; for k algebraically closed use (B3). (b) Let $\tau \otimes F' \cong \tau_0 \otimes_{k_1} F'$ with τ_0 over k_1 , of splitting field l/k_1 Galois with group H_0 . Comparing total spaces inside an algebraic closure of F' we may arrange $EF' = lF'$. Since l/k is finite separable and E/k regular, $l \otimes_k E$ is a field and $[lE : E] = [l : k] = |H_0| [k_1 : k]$; as $lE \subseteq EF'$,

$$|H_0| [k_1 : k] |H| = [lE : E][E : F] \leq [EF' : F] = [lF' : F'] [F' : F] \leq |H_0| [F' : F].$$

- (c) $E = L$ is regular over k and $H = G$. □

Remark 10.3. Without regularity $d_0 < |H|$ is possible ($k = \mathbb{Q}$, $\tau = \mathbb{Q}(t)(\sqrt{2})$ is constant, $d_0 = 1$). Proposition 10.2(c) is the general form of Proposition 3.2(c). For k algebraically closed, $\text{ed}^{[d]}(\tau) \geq 1$ for $d < |H|$: the last slot of the profile is the only one understood in general.

Proposition 10.4 (base-field extension bound). *Let k'/k be finite of degree d . Then $\text{ed}_k^{[d]}(\tau) \leq \text{ed}_{k'}(G)$. More precisely, every field factor F' of $F \otimes_k k'$ has $[F' : F] \leq d$ and $\text{ed}_k(\tau \otimes F') \leq \text{ed}_{k'}(\tau \otimes F') \leq \text{ed}_{k'}(G)$.* [proved]

Proof. $F' \supseteq k'$ and $[F' : F] \leq d$. A descent $\tau \otimes F' \cong \tau_0 \otimes_{F_0} F'$ with $k' \subseteq F_0 \subseteq F'$ and $\text{trdeg}_{k'} F_0 \leq \text{ed}_{k'}(G)$ is also a descent over k , and $\text{trdeg}_k F_0 = \text{trdeg}_{k'} F_0$ since k'/k is algebraic. □

Corollary 10.5 (roots-of-unity defect). *Let e be the exponent of G , $\text{char } k \nmid e$, and $d_e = [k(\mu_e) : k]$. For every G -torsor τ over every $F \supseteq k$,*

$$\text{ed}_k^{[[F(\mu_e):F]]}(\tau) \leq \text{ed}_k^{[d_e]}(\tau) \leq \text{ed}_{k(\mu_e)}(G).$$

In particular: (i) if G is abelian, $\text{ed}_k^{[d_e]}(\tau) \leq \text{rank } G$ [19]; (ii) if G is a p -group and $\text{char } k \neq p$, $\text{ed}_k^{[p-1]}(\tau) \leq \text{ed}_k^{[[k(\mu_p):k]]}(\tau) \leq \text{ed}_{k(\mu_p)}(G)$, the minimal dimension of a faithful representation of G over $k(\mu_p)$ [21]. [proved; the values are cited]

Remark 10.6. For τ_{gen} and $F' = K(\mu_e)$ the torsor $\tau_{\text{gen}} \otimes F'$ is the generic torsor over $k(\mu_e)$, so $\text{ed}_{k(\mu_e)}(\tau_{\text{gen}} \otimes K(\mu_e)) = \text{ed}_{k(\mu_e)}(G)$ exactly, but whether ed_k of this torsor equals $\text{ed}_{k(\mu_e)}(G)$ is not known to us in general. [open]

10.3. Sylow divisibility of the jump degrees.

Lemma 10.7 (q -closure). *Let q be a prime, $S \leq \Gamma_F$ a Sylow pro- q subgroup and $F^{(q)} = (F^{\text{sep}})^S$. Then $F^{(q)}$ is a union of finite subextensions of degree prime to q , every finite separable extension of $F^{(q)}$ has q -power degree, and $\text{ed}_k(\tau; q) = \text{ed}_k(\tau \otimes F^{(q)})$ [25, (2.5)]. If F'/F is finite separable with $q \nmid [F' : F]$, then there is an F -embedding $F' \hookrightarrow F^{(q)}$, and hence $\text{ed}_k(\tau \otimes F') \geq \text{ed}_k(\tau; q)$.* [cited; last statement proved]

Proof. $F' \otimes_F F^{(q)}$ is a product of finite separable extensions of $F^{(q)}$, each of q -power degree, whose degrees sum to $[F' : F]$; if none had degree 1 the sum would be divisible by q . So some factor is $F^{(q)}$, giving the embedding, and (B1) gives the inequality. (For a *given* embedding $F' \subseteq F^{\text{sep}}$ the compositum $F'F^{(q)}$ need not be $F^{(q)}$: $F = \mathbb{Q}$, $q = 2$, $F' = \mathbb{Q}(\sqrt[3]{2})$ has $F' \otimes_{\mathbb{Q}} \mathbb{Q}^{(2)} \cong \mathbb{Q}^{(2)} \times \mathbb{Q}^{(2)}(\sqrt{-3})$.) □

Lemma 10.8 (purely inseparable invariance). *Let $\text{char } k = p$, k perfect, G any finite group, F'/F finite purely inseparable. Then $\text{ed}_k(\tau \otimes F') = \text{ed}_k(\tau)$; consequently $\text{ed}_k^{[d]}(\tau)$ is the minimum over separable F'/F of degree $\leq d$.* [proved]

Proof. As in Proposition 3.6, with $\sigma: F^{1/p^m} \rightarrow F$, $x \mapsto x^{p^m}$; the p^m -power map extends to $F^{1/p^m} F^{\text{sep}} \rightarrow F^{\text{sep}}$ and commutes with all automorphisms, so the induced isomorphism $\Gamma_{F^{1/p^m}} \rightarrow \Gamma_F$ is the canonical one and the classifying homomorphism of $\sigma_*(\tau \otimes F^{1/p^m})$ is φ_τ . This recovers Proposition 3.6 without Witt vectors. $\square$

Proposition 10.9. *For every prime q , $\text{ed}_k(\tau_{\text{gen}}; q) = \text{ed}_k(G; q)$. [cited: [25, 23]; sketch below]*

Sketch. “ $\leq$ ” is the definition. For “ $\geq$ ” let $e = \text{ed}_k(\tau_{\text{gen}}; q)$, realised by a prime-to- q extension K'/K and a descent to K_0 , $\text{trdeg}_k K_0 \leq e$; spread out to a G -torsor $Z \rightarrow B$, $\dim B \leq e$, and to a G -equivariant rational map $W' \dashrightarrow Z$ from the normalisation W' of V in $L \otimes_K K'$, finite of degree $[K' : K]$ over V and étale over a dense G -stable open $V_1 \subseteq V$. Given any torsor τ over an infinite F , the twist ${}^\tau V$ is an affine space over F , so a free F -point in ${}^\tau V_1$ exists and gives an equivariant $f: T \rightarrow V_1$; the pullback $T \times_V W' \rightarrow T$ is finite étale of degree $[K' : K]$, hence $\cong T \times_F A$ for a finite étale F -algebra A of that degree. Some field factor F' of A has $q \nmid [F' : F]$ and $T_{F'} \rightarrow W' \dashrightarrow Z$ is equivariant into a free G -variety, so $\text{ed}_k(\tau \otimes F') \leq \dim B \leq e$ by Lemma 2.3 (valid in any characteristic, Theorem 10.13). $\square$

Theorem 10.10 (Sylow divisibility). *Let $\text{char } k = 0$, or k perfect (then use Lemma 10.8). Let τ be a G -torsor over F .*

- (a) *For every prime q and every $j < \text{ed}_k(\tau; q)$, q divides $d_j(\tau)$: every F'/F with $\text{ed}_k(\tau \otimes F') \leq j$ has $q \mid [F' : F]$.*
- (b) *Every prime q with $\text{ed}_k(G; q) > j$ divides $d_j(\tau_{\text{gen}})$; hence $d_j(\tau_{\text{gen}}) \geq \prod_{q: \text{ed}_k(G; q) > j} q$.*
- (c) *For every $d \geq 1$, $\text{ed}_k^{[d]}(\tau_{\text{gen}}) \geq \min_{1 \leq d' \leq d} \max_{q \nmid d'} \text{ed}_k(G; q) \geq \max_{q > d} \text{ed}_k(G; q)$ (q prime).*
- (d) *The prime-to- q profile $\text{ed}_k^{[d]'}(\tau) = \min\{\text{ed}_k(\tau \otimes F') : [F' : F] \leq d, q \nmid [F' : F]\}$ is non-increasing with terminal value $\text{ed}_k(\tau; q)$, reached at a finite prime-to- q degree; for τ_{gen} the terminal value is $\text{ed}_k(G; q)$.*

[proved, granting Proposition 10.9]

Proof. (a) is Lemma 10.7; (b) adds Proposition 10.9; (c) a witness for $\text{ed}^{[d]}$ has some degree $d' \leq d$ and (a) applies to every $q \nmid d'$; (d) is Lemma 10.7 together with the finiteness argument of Proposition 3.7(a). $\square$

Remark 10.11. (i) In the language of Theorem 4.2 and Remark 4.5, (a) is Reichstein–Scavia’s Theorem 10.1 applied to the correspondence of degree prime to q furnished by a free closed point; the proof above via the q -closure is the same mechanism. (ii) Consistency with Sections 3–9: for $G = \mathbb{Z}/p^n$ in characteristic p , $\text{ed}_k(G; p) \leq 1$ and $\text{ed}_k(G; q) = 0$ for $q \neq p$, so Theorem 10.10 only says $p \mid d_0$, consistent with $d_0 = p^n$; it says nothing about $d_1, \dots, d_{n-1}$, which is the point of this note. In characteristic 0 the constraint is much stronger.

Proposition 10.12 (q -groups). *Let G be a q -group of order q^n , $\text{char } k \neq q$ (k perfect if positive), $c = [k(\mu_q) : k]$. Then: (a) $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k(G; q)$ for $c \leq d \leq q - 1$, and $\text{ed}_k(G; q) \leq \text{ed}_k^{[d]}(\tau_{\text{gen}}) \leq \text{ed}_k(G)$ for $d < c$; (b) $\text{ed}_k(G; q) = \text{ed}_{k(\mu_q)}(G)$ is the minimal dimension of a faithful $k(\mu_q)$ -representation [21]; (c) $q \mid d_j(\tau_{\text{gen}})$ for every $j < \text{ed}_k(G; q)$, and $d_0 = q^n$; (d) if $\mu_q \subset k$: $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k(G)$ for $d < q$ and all steps of the profile occur at multiples of q , i.e. $\text{ed}_k^{[d]}(\tau_{\text{gen}}) = \text{ed}_k^{[q \lfloor d/q \rfloor]}(\tau_{\text{gen}})$ for $d \geq q$. [proved, modulo [21] and $\text{ed}(\cdot; q)$ being invariant under prime-to- q extensions of k]*

Proof. Lower bounds: every degree $< q$ is prime to q (Theorem 10.10). Upper bound for $d \geq c$: Corollary 10.5 and $\text{ed}_{k(\mu_q)}(G) = \text{ed}_{k(\mu_q)}(G; q) = \text{ed}_k(G; q)$ [21, 23]. (c) is Theorem 10.10(b) and Proposition 10.2(c); (d) is (a)+(c) with $c = 1$. $\square$

Compare Proposition 3.7: in characteristic p the prime-to- p tail of a p -group collapses to ≤ 1 ; in characteristic $\neq q$ the prime-to- q tail of a q -group is *constant*, equal to $\text{ed}_k(G; q)$.

10.4. Geometric meaning and induced torsors in every characteristic.

Theorem 10.13. *Theorem 4.2 and Corollary 4.4 hold for every finite group G , every field k and every characteristic: $\text{ed}_k^{[d]}(\tau) \leq m$ iff some quasi-projective generically free primitive G -variety Z of dimension $\leq m$ has a closed point of degree $\leq d$ in the free locus of ${}^\tau Z$, iff there are such Z , F'/F of degree $\leq d$ and a G -equivariant morphism $T_{F'} \rightarrow Z^{\text{free}}$. For τ_{gen} this is a G -correspondence $V \rightsquigarrow Z$ of degree $\leq d$ into Z^{free} . [proved]*

Proof. The proof of Theorem 4.2 uses only: the existence of the twist ${}^\tau Z = (T \times_k Z)/G$ as a quasi-projective F -scheme compatible with base change and open immersions [20, §3]; the identification of F -points of ${}^\tau Z$ with equivariant maps $T \rightarrow Z_F$ (elementary for finite G); the fact that $Z^{\text{free}} \rightarrow Z^{\text{free}}/G$ is a G -torsor (SGA 1, V, finite group acting freely); and spreading out. None of these depends on the characteristic or on G being a cyclic p -group. $\square$

Lemma 10.14 (induced torsors). *For finite groups $H \leq G$ and an H -torsor τ_H over F , $\text{ed}_k(\text{Ind}_H^G \tau_H) = \text{ed}_k(\tau_H)$. [proved]*

Proof. “ $\leq$ ”: descend and induce. “ $\geq$ ”: reduce to T_H a field (if $T_H = \text{Ind}_{H'}^H T'$ with T' a field, then $\text{Ind}_H^G \tau_H = \text{Ind}_{H'}^G \tau'$ and $\text{ed}(\tau_H) \leq \text{ed}(\tau')$). Then the argument of Proposition 3.9 goes through for non-abelian G : a descent $T_0 \rightarrow \text{Spec } F_0$ has a connected component T'_0 with stabiliser H_0 ; the components of $T'_0 \otimes_{F_0} F$ are among the gT_H and are permuted transitively by H_0 , so after conjugating H we may assume $T_H \subseteq T'_0 \otimes_{F_0} F$ and $H \leq H_0$. With $F_1 = (T'_0)^H$ and $F_1 \otimes_{F_0} F = \prod F_i$, the factor $T'_0 \otimes_{F_1} F_{i_0}$ containing T_H is an H -torsor over F_{i_0} on whose components H acts transitively; being H -stable, T_H is that factor, and $F_{i_0} = T_H^H = F$. So τ_H descends to F_1 , $\text{trdeg}_k F_1 = \text{trdeg}_k F_0$. $\square$

Corollary 10.15 (subgroup profile). *For every subgroup $H' \leq G$: $\text{ed}_k^{[[G:H']]}(\tau_{\text{gen}}) \leq \text{ed}_k(\tau_{\text{gen}} \otimes L^{H'}) = \text{ed}_k(H')$. Hence $d_j(\tau_{\text{gen}}) \leq \min\{[G:H'] : \text{ed}_k(H') \leq j\}$. [proved]*

Proof. $\tau_{\text{gen}} \otimes L^{H'} = \text{Ind}_{H'}^G(L/L^{H'})$, and $L/L^{H'}$ is the generic H' -torsor of $V|_{H'}$ over k ; apply Lemma 10.14. $\square$

This is the general form of Proposition 3.9 and of the root-field reductions of Example 10.20.

10.5. Examples in characteristic zero.

Example 10.16 ($\mathbb{Z}/n$). Let $\mu_n \subset k$, $\text{char } k \nmid n$. The generic torsor $K(\sqrt[n]{a})/K$, $K = k(a)$, has $\text{ed} = 1$ and $d_0 = n$ (Proposition 10.2), so its profile is $(1, \dots, 1, 0)$. [proved]

Theorem 10.17 ($(\mathbb{Z}/2)^n$). *Let k be algebraically closed of characteristic $\neq 2$, $G = (\mathbb{Z}/2)^n$, $F = K = k(a_1, \dots, a_n)$, $\tau_{\text{gen}} = (a_1, \dots, a_n) \in H^1(F, \mathbb{Z}/2)^n$.*

- (a) $\text{ed}^{[2^j]}(\tau_{\text{gen}}) \leq n - j$ for $0 \leq j \leq n$, and $d_0 = 2^n$.
- (b) Every F'/F of odd degree has $\text{ed}(\tau_{\text{gen}} \otimes F') = n$; all d_j , $j < n$, are even.
- (c) For every quadratic F'/F , $\text{ed}(\tau_{\text{gen}} \otimes F') \geq n - 1$. Hence $\text{ed}^{[2]} = \text{ed}^{[3]} = n - 1$, $d_{n-1} = 2$, and $d_{n-2} = 4$ for $n \geq 2$.

- (d) $\text{ed}^{[d]} = 1$ for $2^{n-1} \leq d < 2^n$. Complete profiles: $(1, 0)$ for $n = 1$; $(2, 1, 1, 0)$ for $n = 2$; $(3, 2, 2, 1, 1, 1, 0)$ for $n = 3$.

[proved; inputs cited]

Proof. (a) Over $F(\sqrt{a_1}, \dots, \sqrt{a_j})$ the class is $(1, \dots, 1, a_{j+1}, \dots, a_n)$. (b) $\text{ed}_k(G; 2) = \text{ed}_k(G) = n$ [19, 21]; apply Theorem 10.10. (d) from (a), (b), (c) and Proposition 10.2.

(c) Write $(a_I) = \bigcup_{i \in I} (a_i) \in H^{|I|}(F, \mathbb{Z}/2)$. Suppose $\text{ed}(\tau_{\text{gen}} \otimes F') \leq n-2$ for $F' = F(\sqrt{f})$. The class descends to F_0 with $\text{trdeg}_k F_0 \leq n-2$, so $H^{n-1}(F_0, \mathbb{Z}/2) = 0$ [26, II.4.2] and $(a_I)_{F'} = 0$ for all $|I| = n-1$. By the Arason exact sequence $H^{r-1}(F) \xrightarrow{(f) \cup} H^r(F) \xrightarrow{\text{res}} H^r(F(\sqrt{f}))$ [17], $(a_{[n] \setminus i}) \in (f) \cup H^{n-2}(F)$ for all i . We show by induction on m that for $F_m = k(a_1, \dots, a_m)$ there is no $g \in F_m^\times$ with $(a_{[m] \setminus i}) \in (g) \cup H^{m-2}(F_m)$ for all $i \in [m]$. For $m = 1$: $(a_\emptyset) = 1 \neq 0$ in H^0 . For $m \geq 2$ pass to $\hat{F} = F_{m-1}((a_m))$, where $H^r(\hat{F}) = H^r(F_{m-1}) \oplus H^{r-1}(F_{m-1}) \cup (a_m)$ and units have $(u) = (\bar{u})$ [26, II.4.3]; the classes (a_I) restrict injectively (their residues are nonzero by induction on the residue maps). Write $g = a_m^\varepsilon u$ and the given $x_i = y_i + z_i \cup (a_m)$. If $\varepsilon = 1$, take $i = m$: comparing the two summands of $(a_{[m-1]}) = (a_m u) \cup x_m$ gives $y_m = (\bar{u}) \cup z_m$ and $(a_{[m-1]}) = (\bar{u}) \cup (\bar{u}) \cup z_m = 0$ (as $-1 \in k^{\times 2}$), a contradiction. If $\varepsilon = 0$, for $i < m$ the second summands give $(a_{[m-1] \setminus i}) = (\bar{u}) \cup z_i$ for all $i \in [m-1]$, contradicting the induction hypothesis with $g = \bar{u}$. So no quadratic extension lowers ed by two; with (a), (b) this gives $\text{ed}^{[2]} = \text{ed}^{[3]} = n-1$, $d_{n-1} = 2$, $d_{n-2} = 4$. $\square$

Remark 10.18. Whether $\text{ed}^{[2^j]}(\tau_{\text{gen}}) = n-j$ for $2 \leq j \leq n-2$ is open; the first case is $n = 4$: is there a quartic extension of $k(a_1, \dots, a_4)$ splitting all six quaternion algebras (a_i, a_j) ? Non-Galois quartics have no kernel description as clean as Arason's, which is what makes $\text{ed}^{[4]}$ hard. [open]

Example 10.19 ($\mathbb{Z}/p$ over $\mathbb{Q}$). Let p be an odd prime and τ any $\mathbb{Z}/p$ -torsor over $F \supseteq \mathbb{Q}$. By Corollary 10.5 and Example 10.16, $\text{ed}_{\mathbb{Q}}^{[F(\mu_p):F]}(\tau) \leq 1$, so $d_1(\tau) \leq p-1$ for every such torsor, although $\text{ed}_{\mathbb{Q}}(\mathbb{Z}/p) \geq 2$ for $p \geq 5$ [22]: $\text{ed}_k(G) = 1$ in characteristic 0 iff $G \hookrightarrow \text{PGL}_2(k)$, and $\mathbb{Z}/p \not\hookrightarrow \text{PGL}_2(\mathbb{Q})$ for $p \geq 5$ since $\zeta_p + \zeta_p^{-1} \notin \mathbb{Q}$. The profile of the generic $\mathbb{Z}/p$ -torsor over $\mathbb{Q}$ thus drops to 1 at a degree dividing $p-1$, prime to p , consistent with Theorem 10.10 since $\text{ed}_{\mathbb{Q}}(\mathbb{Z}/p; p) = 1$. Whether $d_1(\tau_{\text{gen}}) = [\mathbb{Q}(\zeta_p + \zeta_p^{-1}) : \mathbb{Q}] = (p-1)/2$ is open. [proved from [22]]

Example 10.20 (symmetric groups). Let $\text{char } k = 0$, $L = k(x_1, \dots, x_n)$, $F = K = k(s_1, \dots, s_n)$, so τ_{gen} is the generic degree- n étale algebra. For $F_m = F(x_1, \dots, x_m)$, $[F_m : F] = n(n-1) \cdots (n-m+1)$, Corollary 10.15 and (B4) give

$$\text{ed}_k(\tau_{\text{gen}} \otimes F_m) = \text{ed}_k(S_{n-m}), \quad \text{so} \quad \text{ed}_k^{[n(n-1) \cdots (n-m+1)]}(\tau_{\text{gen}}) \leq \text{ed}_k(S_{n-m}).$$

With $\text{ed}_k(S_n; q) = \lfloor n/q \rfloor$ [24], Theorem 10.10 gives $\text{ed}^{[d]}(\tau_{\text{gen}}) \geq \lfloor n/q_d \rfloor$ for the least prime $q_d > d$, so the profile of the generic étale algebra decays no faster than $\sim n/(2d)$, whereas the root-field bound needs degree $\sim n^m$ for a drop of m . For S_6 (with $\text{ed}_k(S_6) = 3$, $\text{ed}_k(S_5) = \text{ed}_k(S_4) = 2$, $\text{ed}_k(S_3) = 1$ [19], and $\text{ed}_k(A_6) = 3$ [cited from memory]): $\text{ed}^{[1]} = 3$; $\text{ed}^{[d]} \geq 2$ for $2 \leq d \leq 5$; $\text{ed}^{[6]} \leq 2$, $\text{ed}^{[120]} \leq 1$, $d_0 = 720$; $2 \mid d_2$, $6 \mid d_1$; so $d_2 \in \{2, 4, 6\}$ and $d_1 \in 6\mathbb{Z} \cap [6, 120]$. The discriminant field L^{A_6} is *not* a witness for $d_2 = 2$, since over it $\text{ed} = \text{ed}_k(A_6) = 3$. [proved from cited inputs]

10.6. Provenance: rewrites versus new statements.

Statement	Status relative to the literature
Prop. 10.1 (shape)	Rewrite of Prop. 3.2 for arbitrary G , k ; formal.
Prop. 10.2 (last jump)	New but elementary; (a) is folklore, (b) the regularity refinement is new.
Prop. 10.4, Cor. 10.5	New (elementary); the values quoted are [19, 21].
Lemma 10.7	Rewrite of [25, (2.5)]; the embedding argument is standard.
Lemma 10.8	Rewrite of Prop. 3.6 in a form valid for every finite G (no Witt vectors).
Prop. 10.9	Known: [25, 23] (ed_q of a versal object).
Thm. 10.10	The mechanism is [25, Thm. 10.1]; the statement about jump degrees and the bound (c) are new.
Prop. 10.12	New corollary of [21].
Thm. 10.13	Rewrite of Thm. 4.2 in full generality; ingredients [20, 25].
Lemma 10.14, Cor. 10.15	Extends Prop. 3.9 to non-abelian G ; probably known to experts, we found no reference.
Ex. 10.16	Folklore.
Thm. 10.17	(a),(b),(d) are immediate from the above; (c), i.e. $\text{ed}^{[2]} = \text{ed}^{[3]} = n - 1$ and $d_{n-2} = 4$, is new (Arason sequence [17] + residue induction).
Ex. 10.19	New observation from [22].
Ex. 10.20	The identity $\text{ed}(\tau_{\text{gen}} \otimes F_m) = \text{ed}(S_{n-m})$ is new but easy; inputs [19, 24].

10.7. Discussion and open questions. In characteristic 0 the profile of τ_{gen} is squeezed between Sylow divisibility (Theorem 10.10: a prime q with $\text{ed}_k(G; q) > j$ divides every d_i , $i \leq j$, and the prime-to- q tail is constant), the roots-of-unity layer (Corollary 10.5: within degree $[k(\mu_e) : k]$, prime to $|G|$ when $|G|$ is a prime power, the profile drops to the representation-theoretic value $\text{ed}_{k(\mu_e)}(G)$), and the subgroup upper bounds (Corollary 10.15). What is unknown is whether non-Galois or non-subgroup extensions can beat the subgroup witnesses. In characteristic p for a p -group (Sections 3–9) there is no Sylow divisibility beyond $p \mid d_0$ and no roots-of-unity layer; the whole profile lives in p -power degrees and records Artin–Schreier–Witt layers. Theorem 10.13 places both regimes in one framework: the profile is the minimal degree of a G -correspondence into a free G -variety of bounded dimension, Theorem 10.10 isolates the part of that degree forced by [25, Thm. 10.1], and the rest, the q -power part in characteristic $\neq q$ and the p -power part in characteristic p , is where the profile carries information beyond ed and $\text{ed}(\cdot; q)$.

Question 10.21. (1) For $(\mathbb{Z}/2)^n$ over an algebraically closed field of characteristic $\neq 2$, is $d_j(\tau_{\text{gen}}) = 2^{n-j}$ for all j (known for $j \in \{0, 1, n-1, n\}$)? For $(\mathbb{Z}/q)^n$ with $\mu_q \subset k$, the cyclic-kernel analogue of the Arason sequence should give $d_{n-1} = q$ [expected]. (2) For $\mathbb{Z}/p$ over $\mathbb{Q}$, is $d_1(\tau_{\text{gen}}) = (p-1)/2$? More generally, when is Corollary 10.5 sharp, and can an extension not of the form Fk' do better than adjoining roots of unity? (3) Determine $d_2(\tau_{\text{gen}}(S_6)) \in \{2, 4, 6\}$; does $d_j(\tau_{\text{gen}}(S_n))$ grow linearly or super-polynomially in n ? (4) Is the profile of $\tau_{\text{gen}}(V)$ independent of the faithful representation V ? (Presumably yes, by specialisation as in [25].)

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